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United States Patent Application Publication 20250269891
Kind Code A1
Publication Date August 28, 2025
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Systems and Methods for a Crate Lifting Mechanism

Abstract

A crate lifting system can include two or more extendable supports configured to slide under a crate and a set of lifters attachable, each of the lifters to be attached to an end of the extendable supports, wherein each lifter is configured to raise the crate from a surface. The crate lifting system can also include a set of bolts that engage with the extendable supports through the set of lifters, wherein the set of bolts are configured to lift the crate as the set of bolts are rotated.

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Family ID: **1000007726348**

Appl. No.: **18/586245**

Filed: **February 23, 2024**

Publication Classification

Int. Cl.: **B62B5/00** (20060101)

U.S. Cl.:

CPC **B62B5/0086** (20130101); **B62B5/0083** (20130101);

Background/Summary

BACKGROUND

[0001] The transportation and handling of heavy objects, such as crates, within industrial, commercial, and logistical environments present significant challenges, particularly when it comes to ensuring the safety of personnel and preventing injury. Traditional methods of moving heavy crates often require manual lifting or the use of forklifts, which may not always be practical or available, especially in confined spaces or when crates are placed against walls.

[0002] Historically, one form of lifting heavy objects involved manual labor. This included techniques such as bending the knees and keeping the back straight to lift with the legs rather than the back, using the human body as a lever. Teams of workers would often lift in unison to distribute the weight of an object more evenly. However, these methods are inherently limited by human strength and are associated with a high risk of injury, particularly to the back and spine.

[0003] To overcome the limitations of human strength, simple mechanical aids were introduced. The lever is one example of a simple machine that has been used. By applying force at one end of a rigid bar that pivots around a fulcrum, a lever can amplify the lifting force and move heavy loads.

[0004] Techniques herein are configured to address the technical challenges of lifting heavy objects by providing a versatile and efficient means of lifting and moving heavy crates without the need for extensive manual labor or specialized machinery such as forklifts. The system is adaptable to various crate orientations and positions, whether there is ample space around the crate or if it is positioned against a wall with limited access.

BRIEF DESCRIPTION

[0005] This section is provided to introduce a selection of concepts that are further described below in the Detailed Description. This Brief Description is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

[0006] In one example, a crate lifting system can include two or more extendable supports configured to slide under a crate and a set of lifters attachable, each of the lifters to be attached to an end of the extendable supports, wherein each lifter is configured to raise the crate from a surface. The crate lifting system can also include a set of bolts that engage with the extendable supports through the set of lifters, wherein the set of bolts are configured to lift the crate as the set of bolts are rotated.

[0007] In one example, a method for lifting a crate can include mounting a set of boots to each end of the crate when extendable support insertion is not possible, attaching a lifter to each of the boots, and raising the crate from a surface using bolts that engage with the each of the boots through the lifter attached to each of the boots.

[0008] In another example, a crate lifting system can include two or more extendable supports configured to slide under a crate, an electronic lifter attachable to each end of the two or more extendable supports, wherein the lifter includes an integrated electric motor for raising and lowering the crate, and a set of rotating wheels attached to each electronic lifter, the set of rotating wheels facilitating movement of the lifted crate. The crate lifting system can also include an emergency stop feature configured to immediately stop a lifting operation for the crate.

[0009] Various other features, objects, and advantages of the invention will be made apparent from the following description taken together with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The present disclosure will be better understood from reading the following description of non-limiting examples, with reference to the attached drawings broadly described below:

[0011] FIG. **1** is a perspective view of a lifting mechanism attached to a crate;

[0012] FIG. **2** is a perspective view of a lifting mechanism attached to an extendable support;

[0013] FIGS. **3**A, **3**B, and **3**C depict components of the lifting mechanism that include extendable supports;

[0014] FIG. **4** depicts components of a lifting mechanism that are attachable to a crate;

[0015] FIGS. **5**A, **5**B, and **5**C are block diagrams of the lifting mechanism components that are attachable to a crate;
[0016] FIGS. **6**A, **6**B, and **6**C are block diagrams of the lifting mechanism components that are attachable to an extendable support;
[0017] FIG. **7** is a process flow diagram for operating a lifting mechanism with extendable supports;
[0018] FIG. **8** is a process flow diagram for operating a lifting mechanism that is attachable to a crate;
[0019] FIG. **9** is a perspective view of a lifting mechanism coupled to a motorized component;
[0020] FIG. **10** is a block diagram of a device for operating a lifting mechanism; and
[0021] FIG. **11** is a block diagram of a computer-readable media for operating a lifting mechanism.
[0022] The drawings illustrate specific aspects of the described components, systems and methods for a lifting mechanism. Together with the following description, the drawings demonstrate and explain the principles of the structures, methods, and principles described herein. In the drawings, the thickness and size of components may be exaggerated or otherwise modified for clarity. Well-known structures, materials, or operations are not shown or described detail to avoid obscuring aspects of the described components, systems and methods.

DETAILED DESCRIPTION

[0023] Embodiments of the present disclosure will now be described, by way of example, with reference to FIGS. **1-11**, in which the following description relates to various examples of a crate lifting mechanism described herein. The following description relates to various techniques, methods, non-transitory computer-readable media, and systems for lifting crates or any other suitable objects with a lifting mechanism.
[0024] Techniques herein have the technical advantage of lifting crates or any other suitable heavy object from a floor to a higher location for transportation. In some examples, the technical advantages described herein enable lifting heavy crates and other objects even when the crates reside against or in close proximity to a wall.
[0025] In one example, a lifting mechanism can include a set of extendable supports that can slide under a crate, allowing for the attachment of a lifter component (also referred to herein as a “lifter”) at each end of the extendable supports. The lifter mechanism, which can be adjustable in length, is used in conjunction with bolts or other fasteners to raise a crate off the ground. Once lifted, the crate is supported by rotating wheels attached to the lifter component, enabling easy movement.
[0026] In another example, systems herein can be utilized when a crate is placed against a wall and the insertion of extendable supports is not feasible. In this example, the system can include a set of boots that are mounted to each end of the crate. A boot, as referred to herein, can include any suitable attachment that can be affixed directly to a crate, wherein the attachment enables coupling a lifter component to the crate. For example, a lifter component can be mounted to each of the boots and the crate can be lifted by rotating bolts through each lifter. In this example, crates can be moved even when initial placement restricts the use of extendable supports. The techniques herein include an innovative design that simplifies the process of moving heavy crates, reduces the risk of injury, and increasing efficiency in environments where space constraints and safety are concerns.
[0027] FIG. **1** depicts an example block diagram of a lifting mechanism attached to a crate. In some examples, the lifting mechanism **100** can move, lift, or otherwise relocate any suitable object including crates **102**, or the like. The lifting mechanism **100** can be configured to be attached to a crate **102** so that the crate **102** can be lifted, handled, and transported with minimal manual effort and increased safety. The lifting mechanism **100** herein is engineered to be robust yet versatile, capable of securely attaching to various crate sizes and weights.
[0028] In some examples, any number of boots **104** can be coupled or attached to a lifter component **106**. The boots **104** of FIG. **1** reside within the lifter component **106** so that the boots **104** are not visible. The boots **104** attach to the crate **102** and are described in greater detail below

in relation to FIG. **4**.

[0029] The lifter component **106** can include any number of wheels **108** or any other component that enables moving the crate **102**. In some examples, the lifter component **106** can be attached or coupled to the boots **104** with an end bolt **110**. A second top bolt **112** can be placed through the lifter component **106** to affix or attach the lifter component **106** to the boots **104**. Rotating the top bolt **112** can result in the boots **104** rising from the floor or any other surface supporting the crate **102**. A predetermined number of rotations of the top bolt **112** can cause the boots **104** to be lifted to an expected height above the floor. The crate **102** can then be moved in any suitable direction by applying force to the crate **102** in a parallel direction to the floor so that the wheels **108** engage and move or relocate the crate **102**.

[0030] It is to be understood that the block diagram of FIG. **1** is not intended to indicate that the lifting mechanism **100** is to include all of the components shown in FIG. **1**. Rather, the lifting mechanism **100** can include fewer or additional components not illustrated in FIG. **1** (e.g., additional supports, bolts, or boots, etc.).

[0031] FIG. **2** depicts an example block diagram of a lifting mechanism with extendable supports. In some examples, the lifting mechanism **200** can include any number of extendable supports **202** that can be inserted, placed, mounted, or the like underneath a crate **204** or any other suitable object to be lifted. The extendable supports **202** can be coupled or attached to a lifter component **206** on either end of each extendable support **202**. The lifter component **206** can include any number of wheels **208** or any other component that enables moving the crate **204**. In some examples, the lifter component **206** can be attached or coupled to the extendable support **202** with an end bolt **209**. A second top bolt **210** can be placed through the lifter component **206** to affix or attach the lifter component **206** to the extendable support **202**. Rotating the top bolt **210** can result in the extendable support **202** rising from the floor or any other surface supporting the crate **204**. A predetermined number of rotations of the top bolt **210** can cause the extendable support **202** to be lifted an expected amount from the floor. The crate **204** can then be moved in any suitable direction by applying force to the crate **204** in a parallel direction to the floor so that the wheels **208** engage and move or relocate the crate **204**.

[0032] To facilitate the movement of the crate **204** once lifted, the lifting mechanism **200** may be equipped with any suitable number of wheels **208** or casters. These wheels **208** can be omnidirectional to allow for easy maneuvering in tight spaces. In some designs, the wheels **208** may retract or fold away for transporting the lifting mechanism **200** when there is no load on the lifting mechanism **200**.

[0033] It is to be understood that the block diagram of FIG. **2** is not intended to indicate that the lifting mechanism **200** is to include all of the components shown in FIG. **2**. Rather, the lifting mechanism **200** can include fewer or additional components not illustrated in FIG. **2** (e.g., additional extendable supports, bolts, or boots, etc.). For example, the lifting mechanism **200** can include any number of additional features. As discussed in greater detail below in relation to FIG. **9**, the additional features can include load sensors that prevent the lifting mechanism **200** from operating if a crate **204** exceeds a safe weight limit, as well as locking mechanisms that secure the crate in place once lifted. In some examples, a surface of the extendable supports **202** can be fitted with a non-slip section, such as short spikes approximately 1-2 mm in height, among other non-slip features, that can penetrate a wooden crate once the lifting process starts, which can prevent the crate from moving on the extendable supports **202** positioned under the crate. Emergency stop buttons and fail-safes, such as mechanical locks that engage in the event of a malfunction, can also be incorporated into the lifting mechanism **200**.

[0034] In some examples, the lifting mechanism **200** can also include any suitable control system. For example, a control system of the lifting mechanism **200** can be manual, semi-automatic, or fully automatic. Manual controls may involve hand cranks or levers to activate the lifting process. Semi-automatic systems may use buttons or switches to control actuators in the lifting mechanism

200, while fully automatic systems might incorporate sensors and programmable logic controllers (PLCs) to manage the lifting operation.

[0035] FIGS. **3**A, **3**B, and **3**C depict components of the lifting mechanism that include extendable supports. In FIG. **3**A, the lifting mechanism **300** can include a center portion **302** of an extendable support **202** that is illustrated in FIG. **2**. In some examples, the center portion **302** of the extendable support **202** can include any number of slots **304** or openings that enable the center portion **302** of the extendable support **202** to be connected to end supports (**306** of FIG. **3**B). For example, the center portion **302** of the extendable support **202** can include one or more openings **304** or slots that enable a fastener (not depicted) to attach an end support (not depicted) to each side of the center portion **302**. In some examples, the slots **304** can be oval, circular, rectangular, or any other suitable shape and the length of the slots **304** can be based on a length of extension for the extendable support **202**. For example, a slot **304** with a length of 0.5 meters can enable adjusting the extendable support **202** up to 0.5 meters on each side of the extendable support **202**. The slots **304** can enable lifting crates of varying sizes in some examples.

[0036] In some examples, the center portion **302** of the extendable support **202** can enable telescopic movement so that the extendable supports **202** can extend and retract to accommodate crates of various sizes. For examples, the center portion **302** can enable an end support (**306** of FIG. **3**B) to connect at any suitable location to the center portion **302** to allow for the end support (**306** of FIG. **3**B) of the lifting mechanism **300** to extend farther away from the center portion **302** to accommodate for larger crates. In some examples, a size of the openings or slots **304** in the center portion **302** of the lifting mechanism **300** can be configured to determine a distance that the end supports (**306** of FIG. **3**B) can extend away from the center portion **302**. For example, larger slots or openings **304** in the center portion **302** of the extendable supports **202** can enable the extendable supports **202** to be located farther apart and to support larger crates, while smaller slots or openings **3046** in the center portion **302** can reduce the distance between the end supports **3** (**306** of FIG. **3**B) and accommodate smaller crates.

[0037] FIG. **3**B depicts end supports **306** that connect to the center portion **302** of the extendable supports **202**. In some examples, the end supports **306** can include any number of slots or openings **308** for attaching the end supports **306** to the center portion **302** of the extendable support **202**. The end supports **306** can also include any number of bolts **310** that extend away from the end supports **306**. The bolts **310** can be used to attach a lifter component (**312** of FIG. **3**C) to the end support **306**.

[0038] FIG. **3**C depicts an example lifter component **312** that is coupled to the end supports **306**. The lifter component **312** can include a slot **314** through which the bolt **310** of the end support **306** can be placed. In some examples, a wingnut (not depicted) or any other suitable nut can be placed on the bolt **310** extending through the lifter component **312** to couple the lifter component **312** to an end support **306**. In some examples, any number of openings **316** can be located at a top of the lifter component **312** to accept bolts (**310** of FIG. **2**) or other fasteners. For example, one or more bolts (also referred to herein as top bolts) **210** can be rotated or tightened through the top of the lifter component **312** to pull the end support **306** higher and thereby lift a crate off a floor or other surface.

[0039] In some examples, wheels **318** coupled to the lifting mechanism **300** can be configured to rotate any number of degrees without contacting the lifter component **312**. For example, the wheels **318** can be configured so that the wheels **318** are placed farther away from the base plate **320** of the lifter component **312** and the distance between the wheels **318** and the lifter component **312** can be configured based on a radius or diameter of the wheels **318** so that larger wheels **318** with a higher load capacity can be positioned farther from the base plate **320** of the lifter component **312**.

[0040] FIG. **4** depicts components of a lifting mechanism that are attachable to a crate. In FIG. **4**, the lifting mechanism **400** can include an example boot **402** that can include a bolt housing **404** that extends away from a base plate **406** of the boot **402**. The bolt housing **404** can accept any suitable bolt **410** or fastener. In some examples, the bolt **410** can be rotatable to lift the boot **402** within a lifter component **312** illustrated in FIG. **3**C.

[0041] In some examples, a lifting mechanism **400** can utilize any number of boots **402** coupled to lifter components, such as the lifter component **312** of FIG. **3**C For example, the lifting mechanism **400** can include one, two, three, or more boots **402** attached to an edge or side of a crate to be lifted. In some examples, any number of boots **402** can be attached to any number of sides or edges of a crate. In some examples, a combination of boots **402** and any number of extendable supports, such as extendable supports **202** of FIGS. **3**A-**3**C, can be used to lift a crate. For example, any number of extendable supports, such as extendable supports **202** of FIG. **2** can be placed under the crate, in some examples, to provide additional weightlifting capacity.

[0042] In some examples, each boot **402** can be affixed to a crate including any suitable electro-magnet. For example, rather than using screws, bolts, or the like to affix a boot **402** to a crate, a base plate **406** of the boot **402** can be coupled to a magnet or include a magnet. The electro-magnet of the boot **402** can be used to attach the boot **402** to a metal crate. In some examples, a strength of the magnetic force of the magnet of the boot **402** can determine a weight capacity of the boot **402** attached to a lifter component. For example, a magnet that can support 50 kilograms can be coupled to or incorporated into a boot **402** so that the boot **402**, when attached to a lifter component, can support up to 50 kilograms. In some examples, a configuration of a set of boots **402** attached to a crate can lift 100 kilograms or 200 kilograms, among others.

[0043] In some examples, the bolt housing **404** of the boot **402** can be configured to be placed along an edge of a base plate **406** of the boot **402**. The location of the bolt housing **404** at the edge of the boot plate **406**, rather than the center of the boot plate **406**, can prevent incorrect installation of the boot **402** to a crate.

[0044] As discussed above, the boot **402** can be coupled or attached to a lifter component, such as the lifter component **312** of FIG. **3**C, which can include any number of wheels **318**, such as a single wheel, two wheels, three wheels, or the like. The wheels **318** can be fixed or can be configured to rotate any number of degrees. In some examples, the number of wheels **318**, the materials of the wheels **318**, or a combination thereof can determine a weight capacity of the lifting mechanism **400**.

[0045] FIGS. **5**A, **5**B, and **5**C are block diagrams of the lifting mechanism components that are attachable to a crate. FIG. **5**A depicts attaching a boot **402** to a crate with any number of screws or any other fasteners. The boot **402** includes a bolt **410** extending away from the boot **402** that enables securing a lifter component **312** to the boot **402**.

[0046] In some examples, a bolt **410** extending from the boot **402** can accept a wingnut (**502** of FIG. **5**B) to couple a lifter component **312** to the boot **402**. The bolt **410** extending from the boot **402** can include a shank **504** at the base of the bolt **410** to prevent the wingnut **502** from being overtightened, binding the boot **402** and lifter component **312**, and causing friction during the lifting action. The shank **504** can ensure that the lifter component **312** is configured to be raised or lowered as the top bolt **410** is secured to the lifter component **312**.

[0047] FIG. **5**B depicts attaching a lifter component **312** to a boot **402** with a wingnut **502** placed on the bolt **410** extending away from the boot **402**. In some examples, the wingnut **502** may not completely or fully fasten the lifter component **312** to prevent binding to the boot **402**, which enables the bolt **410** extending through the lifter component **312** to raise or lower as a top bolt, such as top bolt **210** of FIG. **2**, is inserted into the lifter component **312** as illustrated in FIG. **5**C.

[0048] In FIG. **5**C, the top bolt **410** can be inserted through the lifter component **312** into the boot **402**. The top bolt **210** can be rotated with any suitable socket, wrench, motorized lifter component, or the like. The top bolt **210** can pull the boot **402** towards the top of the lifter component **312** as the top bolt **210** is rotated, which can result in the crate lifting off the floor or any other surface supporting the crate. In some examples, the top bolt **210** can be rotated any number of times until the crate is lifted off the floor to a predetermined height. In some examples, any number of boots

402 can be fastened or attached to a crate and each boot **402** can be coupled to a lifter component **312** and a top bolt **210** to lift a crate off a floor.

[0049] FIGS. **6**A, **6**B, and **6**C are block diagrams of the lifting mechanism components that are attachable to an extendable support. FIG. **6**A depicts sliding, inserting, or otherwise placing any number of extendable supports **202** under a crate. In some examples, any number of extendable supports **202** can be used to lift the crate off a floor. Each extendable support **202** can include a bolt, such as the end bolt **209** of FIG. **2**, that extends away from the crate and enables attaching a lifter component **312** to each end of an extendable support **202**.

[0050] FIG. **6**B depicts attaching a lifter component **312** to an extendable support **202** with a wingnut **502** placed on the bolt **209** extending away from the extendable support **202**. In some examples, the wingnut **502** may not completely or fully fasten the lifter to the support, which enables the bolt **209** extending through the lifter component **312** to raise or lower as a top bolt **210** is inserted into the lifter component **312** in FIG. **6**C.

[0051] In FIG. **6**C, the top bolt **210** can be inserted through the lifter component **312** into the extendable support **202**. The top bolt **210** can be rotated with any suitable socket, wrench, motorized lifter component, or the like. The top bolt **210** can pull the extendable support **202** towards the top of the lifter component **312** as the top bolt **210** is rotated, which can result in the crate lifting off the floor or any other surface supporting the crate. In some examples, the top bolt **210** can be rotated any number of times until the crate is lifted off the floor to a predetermined height. In some examples, any number of extendable supports **202** can be used to lift a crate.

[0052] FIG. **7** is a process flow diagram for operating a lifting mechanism with extendable supports. In some examples, the method **700** can be implemented with any suitable device such as the lifting mechanisms **300**, **400**, or **900** of FIGS. **3**, **4**, and **9** respectively.

[0053] At block **702**, the method **700** can include placing any suitable number of extendable supports, such as metal supports, or supports of any suitable solid material, under the crate if the crate is in an accessible location. For example, any number of extendable supports can be placed under the crate as long as the crate is not against a wall or other surface that prevents sliding or placing the extendable supports under the crate. In some examples, the supports can be slid under the crate into position at any suitable distance. For example, the extendable supports can be spaced apart at any suitable distance so that the weight of the crate is centered and stable on the supports.

[0054] At block **704**, the method **700** can include attaching a lifter component to each end of each extendable support. For example, the lifter component can be coupled to an extendable support with a bolt, such as end bolt **209** of FIG. **2**, extending from the end of the extendable support. In some examples, a wingnut or any other suitable fastener can be used to secure the bolt within a slot of a side of the lifter component.

[0055] At block **706**, the method **700** can include using any suitable technique for rotating or turning a top bolt into each extendable support through a lifter component. In some examples, a socket and wrench, cordless screwdriver, automated electronic component, or the like can be used to rotate the top bolt and lift the crate off the floor. As the top bolts turn, the crate is gradually lifted off the ground.

[0056] At block **708**, the method **700** can include adjusting a height of the crate as necessary by rotating the top bolt additionally to raise the crate or loosening the top bolt to lower the crate. At block **710**, the method **700** can include moving or relocating the crate to a desired location once the lifting mechanism has lifted the crate to a predetermined height. In some examples, wheels attached or coupled to each lifter component can be used once the crate is elevated to the desired height. These wheels allow the user to move the crate effortlessly within a space. In some examples, as discussed below in relation to FIG. **9**, a remote control component can be used to command the wheels to rotate and swivel as needed.

[0057] The process flow diagram of method **700** of FIG. **7** is not intended to indicate that all of the operations of blocks **702-710** of the method **700** are to be included in every example. Additionally, the process flow diagram of method **700** of FIG. **7** describes a possible order of executing operations. However, it is to be understood that the operations of the method **700** can be implemented in various orders or sequences. In addition, in some examples, the method **700** can also include fewer or additional operations.

[0058] FIG. **8** is a block diagram of the lifting mechanism components that are attachable to a crate. In some examples, the method **700** can be implemented with any suitable device such as the lifting mechanisms **300**, **400**, or **900** of FIGS. **3**, **4**, and **9** respectively.

[0059] At block **802**, the method **800** can include mounting any number of boots to a crate. In some examples, boots can be affixed or mounted to a crate if the crate is against a wall and an extendable support cannot be placed under the crate. In some examples, any number of boots can also be attached to a crate if the boots are preferred over lifting the crate with extendable supports.

[0060] At block **804**, the method **800** can include attaching a lifter component to each boot with a top bolt as illustrated in FIG. **5**B. At block **806**, the method **800** can include lifting the crate off a floor by rotating the top bolt through the lifter component. At block **808**, the method **800** can include moving or relocating the crate. In some examples, moving the crate can include applying a force parallel to the floor to engage the wheels of the lifting mechanism. The force can be applied manually or by any suitable electronic component coupled to the crate and the lifting mechanism.

[0061] The process flow diagram of method **800** of FIG. **8** is not intended to indicate that all of the operations of blocks **802-808** of the method **800** are to be included in every example. Additionally, the process flow diagram of method **800** of FIG. **8** describes a possible order of executing operations. However, it is to be understood that the operations of the method **800** can be implemented in various orders or sequences. In addition, in some examples, the method **800** can also include fewer or additional operations.

[0062] FIG. **9** is a perspective view of a lifting mechanism coupled to a motorized component. In some examples, the lifting mechanism **900** can include any number of lifter components **312** coupled to any number of motorized components **902**. The motorized components **902** can be configured to automatically operate the lifter components **312** by automatically rotating top bolts **210** in the lifter components **312** or any other suitable components. The motorized components **902** can automatically operate the rotation of top bolts **210** into boots **402** of FIG. **4** or extendable supports **202** of FIG. **2**.

[0063] In some examples, each lifter component **312** can be coupled or attached to a single motorized component **902** or each lifter component **312** can be coupled or attached to a shared motorized component **902**. As discussed above in relation to FIGS. **1-8**, lifting mechanism **900** can include any number of extendable supports, boots, lifter components, sensors to indicate the crate has reached a preset height off the floor and the like. The extendable supports, boots, and lifter components can be made from any suitable material such as steel or plastic, among others.

[0064] In some examples, each motorized component **902** can be coupled to a power source **904** that can be included in the lifting mechanism **900** or the power source **904** can be externally located. For example, the power source **904** can be a rechargeable battery pack or a direct connection to an electrical outlet. In some examples, the power source **904** can provide power to a control panel **906**. The control panel **906** can remotely control one or more motorized components **902** by executing instructions that determine when to rotate a top bolt **210** of the lifting mechanism **900**, how much to rotate a top bolt **210** of the lifting mechanism **900**, a preset distance between a crate and a surface to be maintained, or the like. In some examples, a control panel **906** can control or manage multiple lifter components **312** to ensure that a crate is level in the lifting mechanism **900**.

[0065] In some examples, the control panel **906** can be controlled or receive instructions from a display panel **908**, a remote control unit **910**, or the like. The display panel **908** can display real-time information for the lifting mechanism **900** indicating a status of a lift such as current height of a crate, weight load of a crate, battery level, an orientation of the crate in relation to the surface under the crate, and the like. The system may include an audible alarm when moving plus a visual indicator (flashing light) as a warning that the crate may move at any moment. In some examples, the display panel **908** can display safety alerts that indicate if a crate is being moved without an expected height from a floor, a crate is being moved that exceeds a predetermined weight, a crate is being moved while the crate in unlevel, or the like. The display panel **908** can also provide an emergency stop button, among other inputs, that can halt operation of the lifting mechanism and apply brakes to the drive wheels of the lifting mechanism **900**.

[0066] In some examples, the lifting mechanism **900** can include activating lifter components **312** using a remote control unit **910**. The remote control unit **910** can send a wireless signal to the motorized components **902**. For example, the motorized components **902** can include a motor (not depicted) that can turn to operate the top bolt **210** within the lifter component **312**. As discussed above, rotation of the top bolts **210** can control raising or lowering the lifting mechanism **900** and the crate coupled to the lifting mechanism **900**. In some examples, the remote control unit **910** can be configured to control any number of motorized components **902** coupled or otherwise attached to multiple crates. Accordingly, the remote control unit **910** can be configured to control a height of multiple crates from a surface or floor.

[0067] In some examples, the lifting mechanism **900** can include any number of safety features. For example, load weight sensors **912** can be coupled to the lifting mechanism or included in the lifting mechanism **900** to ensure that the lifting mechanism **900** is not overloaded, preventing operation if the crate's weight exceeds the safe operating capacity. In some examples, the lifting mechanism **900** can enable moving or lifting crates or other objects that are sensitive or uneven. In some examples, the lifting mechanism **900** can also include any number of proximity sensors **914** that can enable the lifting mechanism **900** to avoid doors, other crates, people, objects, and the like.

[0068] In some examples, the lifting mechanism **900** can also be integrated with other systems. For example, the lifting mechanism **900** may be designed to integrate with other systems, such as warehouse management systems (WMS) or enterprise resource planning (ERP) systems, allowing for coordinated logistics and inventory management. In some examples, incorporating Internet of Things (IoT) technology to connect the lifting mechanism **900** to a computer network can enable tracking, monitoring, and data collection on usage patterns for maintenance scheduling and inventory management.

[0069] The lifting mechanism **900** can also include any number of safety locks that can prevent accidental release or slippage during lifting and moving operations. For example, the lifting mechanism **900** can include an automatic braking system **916** as another vital safety feature. The automatic braking system **916** can engage when the crate is lifted to a certain height or if the remote control unit **910** is not actively issuing commands, preventing the crate from moving unexpectedly. In some examples, any number of the wheels of the lifting mechanism **900** can have a braking system **916** to securely lock the crate in place once it has been moved to the desired location. In some examples, the automatic braking system **916** can also engage when the crate is lifted.

[0070] It is to be understood that the block diagram of FIG. **9** is not intended to indicate that the lifting mechanism **900** is to include all of the components shown in FIG. **9**. Rather, the lifting mechanism **900** can include fewer or additional components not illustrated in FIG. **9** (e.g., additional memory components, embedded controllers, additional modules, additional network interfaces, etc.). In some examples, the lifting mechanism **900** can include any number of attachments for specialized crates or additional functionality, such as calculating a weight measurement of crate, among others. The attachments can include hooks for lifting barrels or straps for securing irregularly shaped items, among others.

[0071] FIG. **10** is a block diagram of an example of a computing device that can operate a crate lifting mechanism. The computing device **1000** may be, for example, a laptop computer, a desktop computer, a tablet computer, or a mobile phone, among others. The computing device **1000** may include a processor **1002** that is adapted to execute stored instructions, as well as a memory device **1004** that stores instructions that are executable by the processor **1002**. The processor **1002** can be a single core processor, a multi-core processor, a computing cluster, or any number of other configurations. The memory device **1004** can include random access memory, read only memory, flash memory, or any other suitable memory systems. The instructions that are executed by the processor **1002** may be used to implement a method that can operate a crate lifting mechanism, as described in greater detail above in relation to FIGS. **2-4** and **6-9**.

[0072] The processor **1002** may also be linked through the system interconnect **1006** (e.g., PCI, PCI-Express, NuBus, etc.) to a display interface **1008** adapted to connect the computing device **1000** to a display device **1010**. The display device **1010** may include a display screen that is a built-in component of the computing device **1000**. The display device **1010** may also include a computer monitor, television, or projector, among others, which is externally connected to the computing device **1000**. The display device **1010** can include light emitting diodes (LEDs), and micro-LEDs, Organic light emitting diode OLED displays, among others.

[0073] The processor **1002** may be connected through a system interconnect **1006** to an input/output (I/O) device interface **1012** adapted to connect the computing device **1000** to one or more I/O devices **1014** The I/O devices **1014** may include, for example, a keyboard and a pointing device, wherein the pointing device may include a touchpad or a touchscreen, among others. The I/O devices **1014** may be built-in components of the computing device **1000** or may be devices that are externally connected to the computing device **1000**.

[0074] In some embodiments, the processor **1002** may also be linked through the system interconnect **1006** to a storage device **1016** that can include a hard drive, an optical drive, a USB flash drive, an array of drives, or any combinations thereof. In some embodiments, the storage device **1016** can include any suitable applications. In some embodiments, the storage device **1016** can include a lifting mechanism manager **1018**. In some embodiments, the lifting mechanism manager **1018** can electronically control a lifting mechanism by rotating any number of bolts or other fasteners coupled or attached to extendable supports under a crate, bolts attached to boots fastened to a crate, or a combination thereof. In some examples, the lifting mechanism manager **1018** can also control an automatic braking system coupled to wheels of a lifting mechanism or a remote control unit that sends instructions to lifting mechanisms. The lifting mechanism manager **1018** can also monitor in real-time load sensing sensors, proximity sensors, or any other sensors coupled or integrated into the lifting mechanism. The lifting mechanism manager **1018** cand also record for storage and investigation/analysis each transport trip by the lifting mechanism. The stored data can be used for future pre-programming of repeat transport trips.

[0075] In some examples, a network interface controller (also referred to herein as a NIC) **1020** may be adapted to connect the computing device **1000** through the system interconnect **1006** to a network **1022**. The network **1022** may be a cellular network, a radio network, a wide area network (WAN), a local area network (LAN), or the Internet, among others. The network **1022** can enable data, such as alerts, among other data, to be transmitted from the computing device **1000** to remote computing devices, remote display devices, and the like. In some examples, the lifting mechanism manager **1018** can transmit, using the NIC **1020** and the network **1022**, an alert to any suitable external device such as a mobile device or a computing device, among others.

[0076] It is to be understood that the block diagram of FIG. **10** is not intended to indicate that the computing device **1000** is to include all of the components shown in FIG. **10**. Rather, the computing device **1000** can include fewer or additional components not illustrated in FIG. **10** (e.g., additional memory components, embedded controllers, additional modules, additional network interfaces, etc.). Furthermore, any of the functionalities of the lifting mechanism manager **1018** may be partially, or entirely, implemented in hardware and/or in the processor **1002**. For example, the functionality may be implemented with an application specific integrated circuit, logic implemented in an embedded controller, or in logic implemented in the processor **1002**, among others. In some embodiments, the functionalities of the lifting mechanism manager **1018** can be implemented with logic, wherein the logic, as referred to herein, can include any suitable hardware (e.g., a processor, among others), software (e.g., an application, among others), firmware, or any suitable combination of hardware, software, and firmware.

[0077] FIG. **11** is an example of a non-transitory machine-readable medium for operating a lifting mechanism, in accordance with examples. The non-transitory, machine-readable medium **1100** can implement the functionalities of the lifting mechanism manager **1018** of FIG. **10**, among others. For example, a processor **1102** can access the non-transitory, machine-readable media **1100**.

[0078] In some examples, the non-transitory, machine-readable medium **1100** can include instructions to execute a lifting mechanism manager **1018**. For example, the non-transitory, machine-readable medium **1100** can include instructions for the lifting mechanism manager **1018** that cause the processor **1102** to operate a lifting mechanism to lift a crate off a floor by rotating any number of bolts or other fasteners. In some examples, the lifting mechanism manager **1018** can execute any instructions described above in relation to FIGS. **1-10**.

[0079] In some examples, the non-transitory, machine-readable medium **1100** can include instructions to implement any combination of the techniques of the lifting mechanism manager **1018** described above.

EXAMPLES

[0080] In one example, a crate lifting system can include two or more extendable supports configured to slide under a crate and a set of lifters attachable, each of the lifters to be attached to an end of the two or more extendable supports, wherein each lifter is configured to raise the crate from a surface. The crate lifting mechanism can also include a set of bolts that engage with the two or more extendable supports through the set of lifters, wherein the set of bolts are configured to lift the crate as the set of bolts are rotated.

[0081] Additionally, or alternatively, the crate lifting system can include a set of rotating wheels attached to each lifter, allowing the lifted crate to be moved. In some examples, the two or more extendable supports enable lifting the crate when there is sufficient space on either side of the crate to insert the two or more extendable supports. Additionally, or alternatively, the set of bolts for raising the crate are rotatable by a socket, a wrench, or a combination thereof. Additionally, or alternatively, the crate lifting system can include a remote control unit configured to wirelessly operate an electronic lifter, including commands for lifting, lowering, and moving the crate.

[0082] Additionally, or alternatively, the crate lifting system can include a safety mechanism integrated into the electronic lifter, the safety mechanism configured to automatically engage a braking system in response to detecting the crate is lifted to a predetermined height. Additionally, or alternatively, the crate lifting system can include a safety mechanism integrated into the electronic lifter, the safety mechanism configured to automatically engage a braking system in response to detecting the remote control unit is not actively issuing commands. Additionally, or alternatively, the crate lifting system can include a set of load weight sensors incorporated into the two or more extendable supports or the electronic lifter, the set of load weight sensors configured to prevent operation of the crate lifting system in response to detecting an operating violation.

[0083] In some examples, a method for lifting a crate includes mounting a set of boots to each end of the crate when extendable support insertion is not possible, attaching a lifter to each of the boots, and raising the crate from a surface using bolts that engage with the each of the boots through the lifter attached to each of the boots.

[0084] In some examples, the lifter is attached to one of the boots using a bolt extending from the boot, the bolt orientation to be parallel to the surface. In some example, the lifter comprises a slot for the bolt extending from the boot parallel to the surface, the slot to enable the bolt to move vertically without contacting the lifter; and a second opening for inserting a top bolt. In some examples, rotating the bolts clockwise lifts the crate from the surface.

[0085] In some examples, a crate lifting system can include two or more extendable supports configured to slide under a crate and an electronic lifter attachable to each end of the two or more extendable supports, wherein the electronic lifter includes an integrated electric motor for raising and lowering the crate. The crate lifting system can also include a set of rotating wheels attached to each electronic lifter, the set of rotating wheels facilitating movement of a lifted crate, and an emergency stop feature configured to immediately stop a lifting operation for the crate.

[0086] Additionally, or alternatively, the crate lifting system can include a remote control unit configured to wirelessly operate the electronic lifter, including commands for lifting, lowering, and moving the crate. Additionally, or alternatively, the crate lifting system can include a safety mechanism integrated into the electronic lifter, the safety mechanism configured to automatically engage a braking system in response to detecting the crate is lifted to a predetermined height. Additionally, or alternatively, the crate lifting system can include a safety mechanism integrated into the electronic lifter, the safety mechanism configured to automatically engage a braking system in response to detecting the remote control unit is not actively issuing commands. Additionally, or alternatively, the crate lifting system can include a set of load weight sensors incorporated into the two or more extendable supports or the electronic lifter, the set of load weight sensors configured to prevent operation of the crate lifting system in response to detecting an operating violation.

[0087] In some examples, the operating violation comprises detecting a weight of the crate exceeds a specified safe operating capacity. In some examples, the operating violation comprises in response to detecting the crate has been lifted more than a predetermined height. In some examples, the emergency stop feature is accessible on the remote control unit or a control panel of the electronic lifter.

[0088] Various embodiments may be a system, a method, an apparatus or a computer program product at any possible technical detail level of integration. The computer program product can include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of various embodiments. The computer readable storage medium can be a tangible device that can retain and store instructions for use by an instruction execution device. The computer readable storage medium can be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific examples of the computer readable storage medium can also include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device such as punch-cards or raised structures in a groove having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

[0089] Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network or a wireless network. The network can comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device. Computer readable program instructions for carrying out operations of various embodiments can be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, configuration data for integrated circuitry, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Smalltalk, C++, or the like, and procedural programming languages, such as the “C” programming language or similar programming languages. The computer readable program instructions can execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer can be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection can be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) can execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform various aspects.

[0090] Various aspects are described herein with reference to flowchart illustrations or block diagrams of methods, apparatus (systems), and computer program products according to various embodiments. It will be understood that each block of the flowchart illustrations or block diagrams, and combinations of blocks in the flowchart illustrations or block diagrams, can be implemented by computer readable program instructions. These computer readable program instructions can be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart or block diagram block or blocks. These computer readable program instructions can also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart or block diagram block or blocks. The computer readable program instructions can also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational acts to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart or block diagram block or blocks.

[0091] The flowcharts and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the flowchart or block diagrams can represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the blocks can occur out of the order noted in the Figures. For example, two blocks shown in succession can, in fact, be executed substantially concurrently, or the blocks can sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams or flowchart illustration, and combinations of blocks in the block diagrams or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

[0092] While the subject matter has been described above in the general context of computer-executable instructions of a computer program product that runs on a computer or computers, those skilled in the art will recognize that this disclosure also can or can be implemented in combination with other program modules. Generally, program modules include routines, programs, components, data structures, etc. that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that various aspects can be practiced with other computer system configurations, including single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as computers, hand-held computing devices (e.g., PDA, phone), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments in which tasks are performed by remote processing devices that are linked through a communications network. However, some, if not all aspects of this disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0093] As used in this application, the terms “component,” “system,” “platform,” “interface,” and the like, can refer to or can include a computer-related entity or an entity related to an operational machine with one or more specific functionalities. The entities disclosed herein can be either hardware, a combination of hardware and software, software, or software in execution. For example, a component can be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, or a computer. By way of illustration, both an application running on a server and the server can be a component. One or more components can reside within a process or thread of execution and a component can be localized on one computer or distributed between two or more computers. In another example, respective components can execute from various computer readable media having various data structures stored thereon. The components can communicate via local or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor. In such a case, the processor can be internal or external to the apparatus and can execute at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, wherein the electronic components can include a processor or other means to execute software or firmware that confers at least in part the functionality of the electronic components. In an aspect, a component can emulate an electronic component via a virtual machine, e.g., within a cloud computing system.

[0094] In addition, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. As used herein, the term “and/or” is intended to have the same meaning as “or.” Moreover, articles “a” and “an” as used in the subject specification and annexed drawings should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form. As used herein, the terms “example” or “exemplary” are utilized to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as an “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent exemplary structures and techniques known to those of ordinary skill in the art.

[0095] The herein disclosure describes non-limiting examples. For ease of description or explanation, various portions of the herein disclosure utilize the term “each,” “every,” or “all” when discussing various examples. Such usages of the term "each," "every," or "all" are non-limiting. In other words, when the herein disclosure provides a description that is applied to "each," "every," or "all" of some particular object or component, it should be understood that this is a non-limiting example, and it should be further understood that, in various other examples, it can be the case that such description applies to fewer than "each," "every," or "all" of that particular object or component.

[0096] As it is employed in the subject specification, the term "processor" can refer to substantially any computing processing unit or device comprising, but not limited to, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. Further, processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units. In this disclosure, terms such as "store," "storage," "data store," data storage," "database," and substantially any other information storage component relevant to operation and functionality of a component are utilized to refer to "memory components," entities embodied in a "memory," or components comprising a memory. It is to be appreciated that memory or memory components described herein can be either volatile memory or nonvolatile memory, or can include both volatile and nonvolatile memory. By way of illustration, and not limitation, nonvolatile memory can include read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), flash memory, or nonvolatile random access memory (RAM) (e.g., ferroelectric RAM (FeRAM). Volatile memory can include RAM, which can act as external cache memory, for example. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), direct Rambus RAM (DRRAM), direct Rambus dynamic RAM (DRDRAM), and Rambus dynamic RAM (RDRAM). Additionally, the disclosed memory components of systems or computer-implemented methods herein are intended to include, without being limited to including, these and any other suitable types of memory.

[0097] What has been described above include mere examples of systems and computer-implemented methods. It is, of course, not possible to describe every conceivable combination of components or computer-implemented methods for purposes of describing this disclosure, but many further combinations and permutations of this disclosure are possible. Furthermore, to the extent that the terms "includes," "has," "possesses," and the like are used in the detailed description, claims, appendices and drawings such terms are intended to be inclusive in a manner similar to the term "comprising" as "comprising" is interpreted when employed as a transitional word in a claim.

[0098] The descriptions of the various embodiments have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A crate lifting system comprising: two or more extendable supports configured to slide under a crate; a set of lifters attachable, each of the lifters to be attached to an end of the two or more extendable supports, wherein each lifter is configured to raise the crate from a surface; and a set of bolts that engage with the two or more extendable supports through the set of lifters, wherein the set of bolts are configured to lift the crate as the set of bolts are rotated.
2. The crate lifting system of claim 1, further comprising: a set of rotating wheels attached to each lifter, allowing the lifted crate to be moved.
3. The crate lifting system of claim 1, wherein the two or more extendable supports enable lifting the crate when there is sufficient space on either side of the crate to insert the two or more extendable supports.
4. The crate lifting system of claim 1, wherein the set of bolts for raising the crate are rotatable by a socket, a wrench, or a combination thereof.
5. The crate lifting system of claim 1, further comprising a remote control unit configured to wirelessly operate an electronic lifter, including commands for lifting, lowering, and moving the crate.
6. The crate lifting system of claim 5, further comprising a safety mechanism integrated into the electronic lifter, the safety mechanism configured to automatically engage a braking system in response to detecting the crate is lifted to a predetermined height.
7. The crate lifting system of claim 5, further comprising a safety mechanism integrated into the electronic lifter, the safety mechanism configured to automatically engage a braking system in response to detecting the remote control unit is not actively issuing commands.
8. The crate lifting system of claim 5, further comprising a set of load weight sensors incorporated into the two or more extendable supports or the electronic lifter, the set of load weight sensors configured to prevent operation of the crate lifting system in response to detecting an operating violation.
9. A method for lifting a crate comprising: mounting a set of boots to each end of the crate when extendable support insertion is not possible; attaching a lifter to each of the boots; and raising the crate from a surface using bolts that engage with the each of the boots through the lifter attached to each of the boots.
10. The method of claim 9, wherein the lifter is attached to one of the boots using a bolt extending from the boot, an orientation of the bolt to be parallel to the surface.
11. The method of claim 10, wherein the lifter comprises a slot for the bolt extending from the boot parallel to the surface, the slot to enable the bolt to move vertically without contacting the lifter; and a second opening for inserting a top bolt.
12. The method of claim 9, wherein rotating the bolts clockwise lifts the crate from the surface.
13. A crate lifting system comprising: two or more extendable supports configured to slide under a crate; an electronic lifter attachable to each end of the two or more extendable supports, wherein the electronic lifter includes an integrated electric motor for raising and lowering the crate; a set of rotating wheels attached to each electronic lifter, the set of rotating wheels facilitating movement of a lifted crate; and an emergency stop feature configured to immediately stop a lifting operation for the crate.
14. The crate lifting system of claim 13, further comprising a remote control unit configured to wirelessly operate the electronic lifter, including commands for lifting, lowering, and moving the crate.
15. The crate lifting system of claim 14, further comprising a safety mechanism integrated into the electronic lifter, the safety mechanism configured to automatically engage a braking system in response to detecting the crate is lifted to a predetermined height.

16. The crate lifting system of claim 14, further comprising a safety mechanism integrated into the electronic lifter, the safety mechanism configured to automatically engage a braking system in response to detecting the remote control unit is not actively issuing commands.
17. The crate lifting system of claim 13, further comprising a set of load weight sensors incorporated into the two or more extendable supports or the electronic lifter, the set of load weight sensors configured to prevent operation of the crate lifting system in response to detecting an operating violation.
18. The crate lifting system of claim 17, wherein the operating violation comprises detecting a weight of the crate exceeds a specified safe operating capacity.
19. The crate lifting system of claim 17, wherein the operating violation comprises in response to detecting the crate has been lifted more than a predetermined height.
20. The crate lifting system of claim 14, wherein the emergency stop feature is accessible on the remote control unit or a control panel of the electronic lifter.